PRODUCT STRATEGY · TRAVEL · CONVERSATIONAL AI

Say it once. Book it safely.

A voice agent for train and bus booking in Paytm Travel: the customer problem, the strategic bet, the product, and the system design that lets a language model work next to money without ever being trusted with it.

1 turn
from a complete Hinglish sentence to bookable options

0
prices, PNRs or refunds generated by the model, by construction

5 of 5
payment failure patterns resolved inside the conversation

01 Executive summary
· Case study at a glance
02 Working backwards: press release & FAQ
03 Problem diagnosis
04 User personas and insights
05 Strategy and prioritisation
06 Product design
07 Context engineering & the harness
08 Payments, failures and refunds
09 Measurement and experiment
10 Pre-mortem and risks
11 Go-to-market
12 Execution plan
A Appendix: prototype and assumptions

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01 Executive summary

THE ONE-PAGE VERSION

Travel inventory is a commodity and price is a race to the bottom. The durable advantage is in who can book, and how safely they can pay. A voice agent built on Paytm's in-app UPI wins on both.

SITUATION

Every major platform sells the same IRCTC trains and the same buses at near-identical prices. Booking UX across the category is still a multi-screen form, designed for confident, English-first metro users.

COMPLICATION

For Hindi-first and first-time online travellers, the form is the drop-off. After payment, "money deducted, ticket pending" turns the most anxious moment into a support ticket. LLMs can now understand these users, but cannot be trusted with fares or refunds.

RESOLUTION

A voice agent where the model only interprets intent. Prices, PNRs and refunds come from systems of record, every spoken number is verified, and payment is one tap on Paytm's own UPI, with the PIN never leaving the device.

What I am recommending

#	MOVE	WHY THIS, WHY NOW
1	**Ship post-booking voice support first** status, cancellation quote, money-deducted resolution	Lowest risk, clearest ROI (ticket deflection). It builds the agent harness, the eval set and customer trust before the agent touches a new booking.
2	**Pilot voice booking for trains at 1%** Hindi-belt states, Android, Travel home	Biggest upside, highest uncertainty. A narrow slice gives a clean read on conversion with bounded downside.
3	**Treat the harness as the product, not the model**	Models are interchangeable and getting cheaper. Domain guardrails, grounding and evals are what compound, and what risk and legal will sign off on.

≤ 2	≥ base	≤ base	0
median turns to options (target)	search-to-booking conversion vs form	wrong-booking rate (guardrail)	payment credentials reaching logs or a model

DECISION REQUESTED

Approve a two-week, 1% pilot of voice train booking against the existing form, with pre-agreed kill criteria (section 10), and staff post-booking voice support as the first production launch on the same stack.

The case study at a glance

FOUR LENSES

Every product decision in this memo answers one of four questions: who it is for, how it reaches them, what they experience, and how it gets built and proven.

1 · USER PERSONAS

Who is this for, and what job are they hiring it to do?

- Three personas: **family booker** (Hindi-first, Tier-2/3), **busy regular** (metro, price-sensitive), **anxious payer** (post-payment, any segment)
- Jobs-to-be-done, behaviours, frustrations and five insights, each with a "so what"
- **Built:** Hinglish one-sentence booking, read-back before booking, cross-mode cheapest/fastest, in-chat payment recovery

Sections 03, 04

2 · GO-TO-MARKET

How does it reach the first 1% of users, and win them?

- **Beachhead:** Hindi-first Android UPI users in Hindi-belt states, plus existing train bookers who need post-booking help
- Positioning, three message pillars, in-app entry points, launch sequence, adoption funnel, unit economics
- **Built:** Hindi/Hinglish entry copy, demo scenarios, walkthrough video, adoption instrumentation

Sections 05, 11

3 · PRODUCT DESIGN

What does the customer experience, and what keeps it safe?

- Six principles, a turn-by-turn conversation design, Now/Next/Later scope
- Context engineering harness: the LLM understands, tools decide, templates speak
- **Built:** voice + visual canvas, explicit approval, 10-min fare hold, in-app UPI, failure lab, credential guard, grounding validator

Sections 06, 07, 08

4 · EXECUTION

How do we build it, prove it and decide to scale it?

- KPI tree, experiment design with power estimate, kill criteria, pre-mortem
- 13-week plan with exit gates; operating model across engineering, risk and support
- **Built:** working prototype, 27 automated tests, 30-utterance NLU eval, live journey analytics, 2-minute walkthrough

Sections 09, 10, 12

What has already been built

PARAMETER	SHIPPED IN THE PROTOTYPE	EVIDENCE
User personas	One-sentence Hinglish booking; asks only for missing details; read-back with fare and refund rule; cheapest and fastest named across modes	Walkthrough video, scenes 1–3
Go-to-market	"Bol ke book karo" entry copy; six one-tap demo scenarios; funnel and adoption events from first turn to confirmed ticket	Insights panel
Product design	Voice + visual canvas; explicit approval; fare hold; in-app UPI; payment failure lab; 3-way reconciliation; refund quote before cancel	Walkthrough video, scenes 4–6
Execution	Context harness with guards and grounding; Gemini with offline fallback; 27 tests; NLU eval set; measured latency p50/p90	Repo, test suite

02 Working backwards

Written as if on launch day, to force clarity on the customer benefit before the solution. Hypothetical.

PRESS RELEASE · BENGALURU

Paytm Travel now books your train when you just say where you're going

Speak in Hindi, English or Hinglish. Confirm once. Pay with Paytm UPI. Done.

Today Paytm Travel launched voice booking for trains. Instead of picking stations, dates and classes across several screens, travellers can say one sentence, for example "*Mumbai se Nagpur agle Friday, do log, sleeper*", and see the right options straight away.

The assistant asks only for what is missing, names the fastest and the cheapest options, and reads back the train, time, total fare and refund rule before booking anything. Nothing is booked without the traveller saying yes. Payment happens on the Paytm UPI screen, where the PIN is entered as usual and is never spoken.

If a payment goes wrong, the assistant fixes it in the same conversation. When money is debited but no ticket appears, it checks the bank, the booking ledger and the railway system in seconds, then either issues the ticket or starts a full refund.

> *"My mother can book her own ticket to Nagpur now. She just tells it where she's going and checks the price it reads back."*

Illustrative customer quote

Customer FAQ

What if it mishears me?
It always reads back the train, date, travellers and total before booking. Say "no" and nothing happens.

Will it ask for my UPI PIN or OTP?
Never. If you say one, it stops you and does not store it.

Can I cancel by voice?
Yes. It tells you the exact refund first, then cancels only if you confirm.

Internal FAQ

Why not let the LLM answer directly?
A single invented fare or PNR is a trust and regulatory incident. The model interprets; systems of record answer.

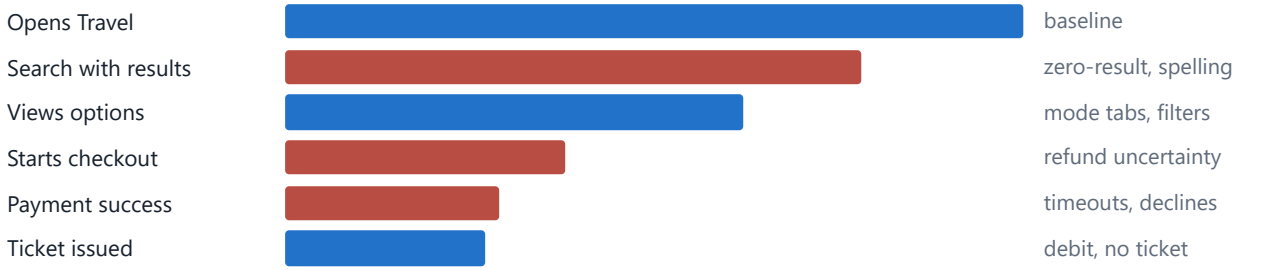
What happens when the model is slow or down?
A deterministic engine with the same output contract takes over mid-conversation. Quality degrades; availability doesn't.

Why trains first?
Hypothesis: the highest share of first-time and Hindi-first bookers, and the most form-heavy flow.

03 Problem diagnosis

The leak isn't one step. It is a form that assumes the customer already knows station codes, class names and fare rules, followed by a payment step that fails with no explanation.

Illustrative booking funnel and where it leaks



Shape for illustration only; red bars mark where I expect the largest leaks. Week one replaces this with real funnel data by city tier and app language.

Root-cause tree

SYMPTOM	ROOT CAUSE	EVIDENCE I WOULD PULL
Abandon at search	Customer thinks in cities and colloquial names ("Bombay", "Jodpur"); the system wants station codes	Zero-result rate; query spelling clusters; app language
Abandon at compare	Train, bus and flight live in separate tabs; the cheaper mode is never seen	Cross-mode views per session; mode switches
Abandon at review	Refund and baggage rules are buried; the fare moves while the user hesitates	Time on review; fare-change errors
Fail at payment	App hand-off, bank timeouts, declines with no clear next step	Failure reasons; retry and recovery rate
Contact support after paying	Callback lost between bank and ledger; no self-serve answer	Top-20 support intents; time to resolution

> **What we know vs what we assume.** Known: the booking form is multi-step, and debit-without-ticket is a recurring failure mode in UPI commerce. Assumed, to validate in week one: which step leaks most for Hindi-first users, and the share of support volume that is payment-related.

04 User personas and insights

WHO STRUGGLES, AND WHAT THEY NEED

Three customers, three jobs

THE FAMILY BOOKER

"Book the right train for my family without asking my son."

45, Wardha. Hindi-first. Books trains a few times a year, often through someone else or an agent. Fears the wrong date more than a higher price.

THE BUSY REGULAR

"Cheapest decent option to Goa, now."

29, Bengaluru. Compares across apps and tabs. Abandons when filters reset. Values speed and seeing all modes at once.

THE ANXIOUS PAYER

"My money went. Where is my ticket?"

Any segment, right after paying. Raises a ticket, waits, may post publicly. Highest churn risk in the journey.

Insights that shape the product

1. **People already describe trips in one sentence.** Origin, destination, date, headcount and mode arrive together, in mixed language. *So what:* extract every slot from one utterance, and never re-ask what was said.
2. **Fear of a wrong booking outweighs price sensitivity for first-time bookers.** ASSUMPTION *So what:* a mandatory read-back with explicit "yes" is a conversion feature, not friction.
3. **Price-sensitive users rarely see the cheaper mode.** ASSUMPTION *So what:* every result set names both the fastest and the cheapest option across train, bus and flight.
4. **After payment, silence is the enemy.** A clear "no money was taken" or "your refund has started" prevents most follow-up contacts. *So what:* payment failures are handled in-conversation, with a specific next step.
5. **Trust is earned by what the assistant refuses to do.** Never asking for an OTP, never booking without consent, never guessing a fare. *So what:* the guardrails are visible in the experience, not hidden in the backend.

What each persona gets

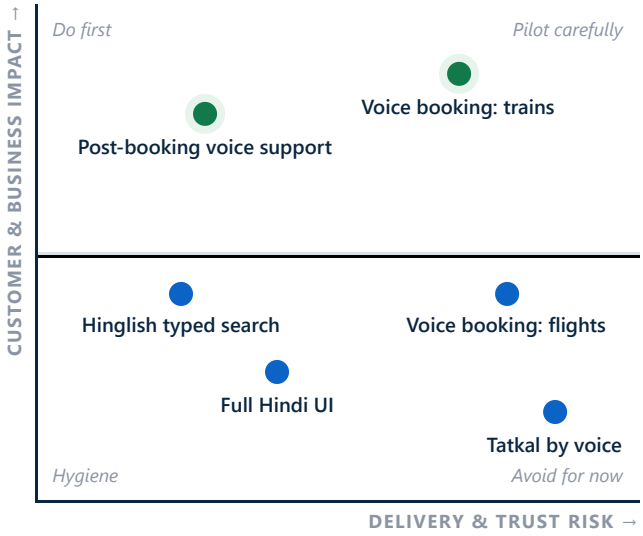
PERSONA	PAIN WE DESIGN AGAINST	WHAT WE BUILT FOR THEM
Family booker	Fear of booking the wrong date; unfamiliar station codes	Hindi/Hinglish one-sentence input, colloquial city names, read-back + explicit "yes"
Busy regular	Comparing modes across tabs; filters resetting	Cross-mode results naming fastest and cheapest; "sabse sasta?", "one day later?" without restarting
Anxious payer	Money debited, no ticket, no answer	In-chat 3-way reconciliation: ticket issued or full auto-refund, with an audit trail

> **Research plan to validate:** 15 interviews in Hindi with Tier-2/3 train bookers (including people who book for others), a review of 200 support transcripts tagged "payment", and search-log mining for colloquial city names and code-switching.

05 Strategy and prioritisation

Paytm's edge isn't inventory. The customer is already inside a UPI app, so payment is one tap rather than a hand-off. The strategy is to win the customers others under-serve, at the step where others fail.

Options against impact and risk



RICE, relative and illustrative

OPTION	R	I	C	E	SCORE
Post-booking voice support	8	2	.8	2	**6.4**
Voice booking: trains	7	3	.5	4	**2.6**
Hinglish typed search	9	1	.7	3	2.1
Full Hindi UI	6	1	.6	3	1.2
Voice booking: flights	4	2	.4	4	0.8

Scores rank options, not forecast outcomes. Reach and effort get replaced with real data before planning.

Why this sequence

Risk before reach

Support conversations reuse 80% of the stack: NLU, harness, payments, evals. They can't book anything wrong.

Learn on real language

Support traffic produces the golden set of real Hindi and Hinglish utterances that booking needs.

Earn trust, then ask for more

Customers who saw the agent fix a payment issue are the most likely early adopters of voice booking.

Deliberately not doing

- **Tatkal by voice:** timing-critical and high-anxiety; failure is expensive. Earn trust first.
- **Autonomous "book it for me":** no booking without explicit, per-transaction consent.
- **A general travel chatbot:** open-ended chat adds cost and risk without a conversion story.

06 Product design

PRINCIPLES, EXPERIENCE, SCOPE

Ask only what's missing
One sentence can fill every field. Confirm what was heard, ask for one thing.

Approval before action
Read back train, time, total and refund rule, then wait for "yes".

The model never states money
Fares, PNRs and refunds come only from systems of record.

Secrets never enter the chat
PIN, OTP and card numbers are blocked before they're logged or modelled.

Every failure has a next step
Failed → retry same order. Debited → reconcile. Cancel → quote first.

Fall back, don't fall over
If the model is slow, a deterministic engine continues the same conversation.

The experience, turn by turn

"Mumbai se Nagpur agle Friday, do log, train"	All five details extracted in one turn. Agent: "*I found 3 options for Fri 20 Nov for 2 travellers. Fastest is the Duronto at 11h 45m for ₹3,700; cheapest is Sleeper for ₹840. Which one should I book?*"
"Sabse sasta wala"	Proposes the cheapest, with departure, duration, total and cancellation rule, and asks "*Shall I book it?*"
"Haan, kar do"	Asks for the lead passenger's name, locks the fare for 10 minutes and opens Paytm UPI. The PIN is entered on the UPI screen only.
"Paise kat gaye, ticket nahi aaya"	Checks bank, ledger and railway inventory, then issues the ticket or starts a full refund, and says which and why.
"Cancel karna hai, kitna wapas milega?"	Quotes the refund (paid − fee per traveller − convenience fee) and cancels only after confirmation.

Scope

HORIZON	INCLUDED
Now	Post-booking voice support · voice booking for trains (SL / 3AC / 2AC, one-way, ≤ 6 travellers) · Hindi, Hinglish, Indian English · in-app UPI · journey instrumentation
Next	Buses · return journeys · streaming speech in and out · saved co-travellers
Later	Flights and hotels · Marathi, Tamil, Bengali · berth preference by voice · Tatkal (only after trust metrics hold)

07 Context engineering & the harness

The model understands, tools decide, templates speak. "Never invent a fare" is enforced in code and tests, not requested in a prompt.

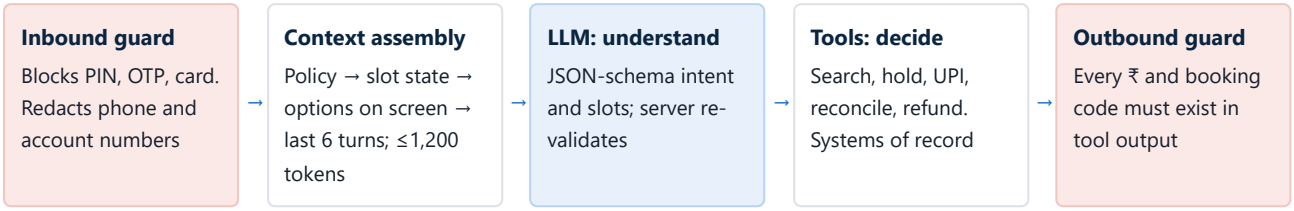


Figure. One conversational turn. Red stages are hard guarantees; the blue stage is the only probabilistic one.

Harness components and the risk each one retires

COMPONENT	BEHAVIOUR	RISK RETIRED
Credential guard	Blocks secrets before storage or model calls; at payment, a bare 4–6 digit number is treated as a PIN	Social-engineering and OTP scams
PII redaction	Masks phone and account numbers before text leaves the device	Data minimisation (DPDP Act 2023)
Budgeted, layered context	Fixed priority order; oldest dialogue trimmed first; policy and state never dropped	Context rot; cost and latency drift
Structured output + validation	Schema-constrained JSON; rejects past dates, implausible headcounts, injection-like text	Malformed or adversarial output
Deterministic overrides	For "yes", "no" and payment complaints, rules beat the model	Probabilistic consent
Slot ledger with provenance	Each detail records value, source and turn	Undebuggable agent behaviour
Grounding validator	Unverified amounts or codes are blocked and logged	Hallucinated fares and PNRs
Graceful degradation	Same output contract from a rules engine on timeout	Vendor outage = product outage

Latency budget

STAGE	TARGET P90	PROTOTYPE
Understand (LLM)	≤ 800 ms	2.1–3.4 s
Tools	≤ 600 ms	0.3–0.5 s
First audio out	≤ 500 ms	~4 s

Measured on the prototype (small sample). Speech and model round-trips dominate, not travel APIs. Levers: streaming TTS, a faster NLU model, cached prompt prefix.

Build vs buy

LAYER	CALL
Language model	Buy; swappable behind one interface
Indic speech	Buy; specialist quality problem
Harness, guardrails, evals	**Build: this is the moat**
Payments, reconciliation	**Build: core competence**

08 Payments, failures, refunds

TRUST LAYER

Most of the product's risk, and much of its upside, sits in the ninety seconds after the customer says "yes".

Resolution logic for "money deducted, ticket pending"

Customer reports a debit without a ticket

Bank / PSP: **not debited**	Bank / PSP: **debited**. Check booking ledger and inventory

Reassure. "Nothing is stuck; a pending debit reverses on its own." Offer retry.	**Seat still held or available** → issue ticket now, against the existing payment.	**Hold expired and seat sold** → full auto-refund, no fees; offer an alternative.

Payment failure patterns

PATTERN	AGENT BEHAVIOUR	PRINCIPLE
Declined in UPI	"No money was taken. Your fare is held for N more minutes. Shall I send it again?"	Reassure first; keep the hold
Bank timeout	Same, naming the cause plainly	Explain, don't blame
Insufficient balance	Same, without judgement in the copy	Dignity in failure
Retry	New request on the **same order id**	Idempotent: no double charge
Cancellation	Quote = paid − fee per traveller (more than 24h vs within 24h of departure) − convenience fee; then confirm	No surprise refunds

Compliance and trust

UPI authentication

PIN entered only on the UPI screen; the assistant never asks for or accepts it. Confirm the in-app intent flow and consent copy with payments.

Failed transactions

Reconciliation and auto-refund paths with an audit trail. Align timelines and messaging with RBI turnaround norms.

Personal data

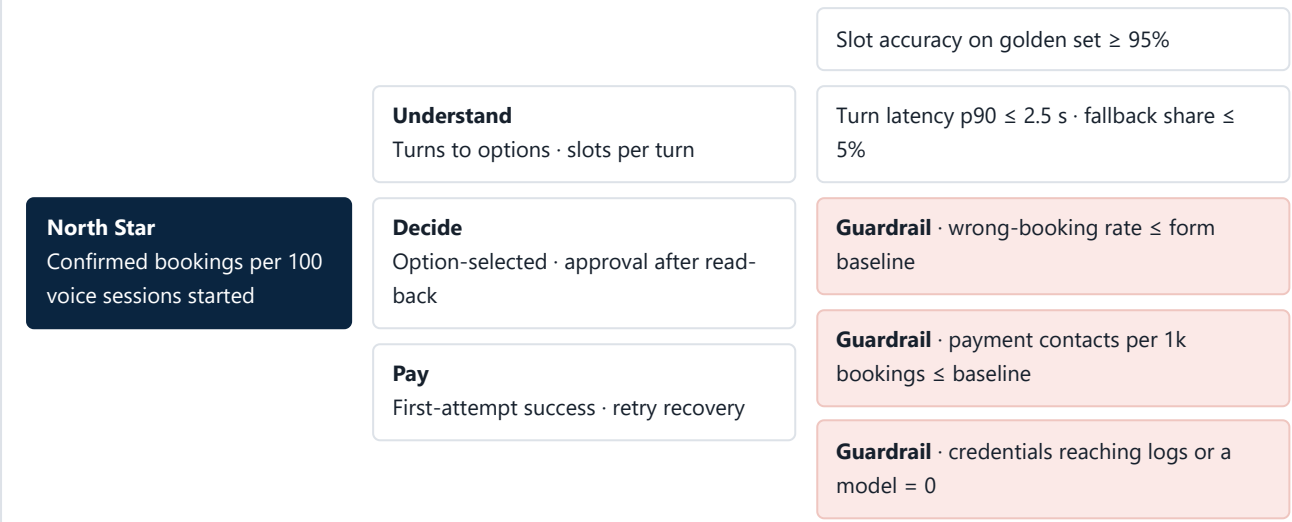
Redaction before the model, no audio stored, keys server-side. Open: consent notice, retention, provider data residency.

Model risk

No generated money values, deterministic consent, per-turn audit logs. Open: model change-management process.

09 Measurement and experiment

HOW WE WILL KNOW



KPI tree: outcome → levers the team moves weekly → quality bars and guardrails.

Experiment design

ELEMENT	DESIGN
Hypothesis	For Hindi-first train bookers on Android, a voice entry point increases search-to-booking conversion without increasing wrong bookings or payment contacts.
Unit and split	User-level randomisation; 1% treatment on Travel home in Hindi-belt states
Primary metric	Search-to-booking conversion. Secondary: time to booking, first-attempt payment success
Power	At an assumed 20% baseline and 10% relative MDE (α 0.05, power 0.8), about 6,500 sessions per arm. Recomputed on the real baseline
Duration	At least two full weeks, to cover weekday and weekend travel intent
Ship rule	Primary not worse *and* every guardrail within bounds; scale to 10% and add buses

Model quality is measured separately from product impact

Offline eval

A golden set of real utterances scored for intent and slot accuracy on every change, for both the model and the fallback.

Online health

Fallback share, latency p50/p90 per stage, ungrounded-claim blocks per 1k turns.

Human review

Weekly review of sampled conversations with support and risk, feeding failures back into the golden set.

10 Pre-mortem and risks

ASSUME IT FAILED. WHY?

It is three months after launch and the pilot was rolled back. The most likely reasons, and what is already in place to prevent each:

FAILURE STORY	SEVERITY	PREVENTION IN THE DESIGN	EARLY WARNING SIGNAL
"It booked the wrong date" after misheard speech	HIGH	Mandatory read-back and explicit approval; slot ledger shows exactly what was understood	Cancellations within 1h tagged "wrong date"
"It told me the wrong fare"	HIGH	Model never states money; grounding validator blocks unverified numbers	Ungrounded-claim blocks above zero
A scam used the assistant's voice to collect an OTP	HIGH	Assistant never asks for credentials and says so; credential guard blocks them	Credential-guard triggers per 1k sessions
"It's too slow, I'll just type"	MEDIUM	Timeouts with instant fallback; latency tracked per stage	Turn p90 above 2.5 s; voice-to-form switches
Cost per booking too high to scale	MEDIUM	1,200-token budget, small JSON outputs, no model "thinking" for NLU	Model cost per confirmed booking
Vendor outage during peak	MEDIUM	Rules engine with the same output contract; one swappable interface	Fallback share above 5%

KILL CRITERIA, AGREED BEFORE LAUNCH

Roll back if wrong-booking rate is more than 20% above the form baseline for 3 consecutive days, if any payment credential reaches logs or a model, or if turn latency p90 exceeds 4 s for more than 24 hours.

Trade-offs I am choosing

Correct over charming

Templated replies feel less magical than free generation. Next to money, predictability beats personality; personality can live in small talk.

Narrow over broad

Trains, Hindi, one-way first. A narrow pilot gives a clean read on one hypothesis; a broad one gives noise and more ways to fail.

11 Go-to-market

REACH, TRUST, FIRST BOOKING

Paytm doesn't need to buy distribution: the customer is already in the app to pay. Go-to-market is about the right entry point, a trust message that lands in Hindi, and a first booking that works.

Beachhead segment

Hindi-first Android users in Hindi-belt states who already pay with Paytm UPI but have never booked travel on Paytm, plus **existing train bookers** who contact support after paying. ASSUMPTION

Positioning

For travellers who find booking forms confusing, Paytm Travel lets you **book a train in one sentence** and pay with Paytm UPI. Unlike other travel apps, it **reads everything back before booking** and **fixes payment problems in the same conversation**.

Message pillars

PILLAR	IN-APP LINE
Simplicity	"Bol ke book karo"
Control	"Aapki 'haan' ke baad hi booking"
Safety	"Hum kabhi PIN ya OTP nahi poochenge"

Entry points and channels

ENTRY POINT	MOMENT	WHY IT CONVERTS
Order details: "Talk to us"	After booking: status, cancellation, money debited	Launches support first; lowest risk, highest intent
Payment-failure screen	Right after a failed or pending UPI payment	Resolves anxiety in the moment it peaks
Mic in Travel search	Search bar with example sentence as placeholder	Zero new navigation; shows what to say
Travel home banner	Hindi-language users, Hindi-belt states	Targets the beachhead only
Seasonal push	Before peak travel periods, once pilot gates pass	Scales demand only after trust is proven

Adoption funnel

STAGE	METRIC
Discover	Entry-point tap rate
Activate	Sessions with ≥1 voice turn
Convert	First confirmed voice booking
Retain	Repeat voice booking in 60 days

Economics and readiness

- **No voice fee:** same convenience fee as the form, so voice never costs the customer more
- **Unit cost:** model + speech cost per confirmed booking tracked against the convenience fee; support deflection counted as savings
- **Readiness:** support trained on agent flows; risk sign-off; vendor SLAs; first-run tip with an example sentence

12 Execution plan

LEARN → BUILD → PROVE → SCALE

WEEKS	PHASE	OUTCOMES	EXIT GATE
1–3	**Learn**	Funnel baselines by city tier and app language; top support intents; 15 Hindi interviews; golden set of 500 real utterances	Baselines agreed with analytics
4–6	**Build the foundation**	Harness on real train search and in-app UPI; streaming speech; post-booking voice support in dogfood	Slot accuracy ≥ 95%; zero credential leaks in red-team
7–10	**Prove**	Support launch to all train bookings; 1% voice-booking experiment; weekly conversation reviews	Primary metric not worse; guardrails hold
11–13	**Decide and scale**	Read-out against pre-agreed gates; 10% plus buses, or stop	Significant lift vs control

Operating model

Engineering
Agree harness interfaces (understand, tools, validate) first, so model and vendor choices stay swappable.

Risk, legal, payments
Sign-off on credential handling, refund messaging and audit logs before any external exposure.

Support and operations
Weekly review of real conversations; agents are the best source of golden-set utterances.

Open questions for the Travel team

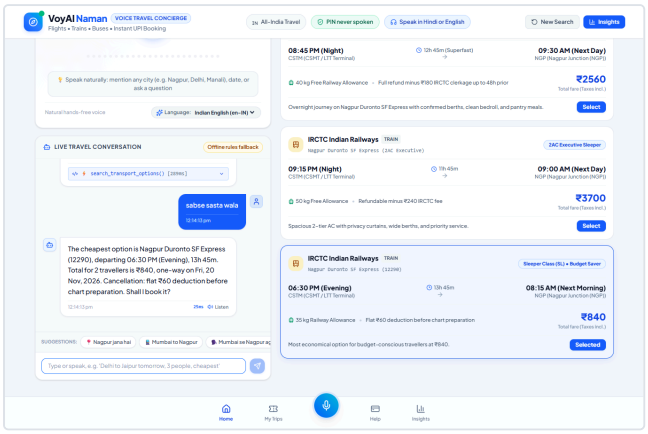
- Which booking step loses the most Tier-2/3 and Hindi-language users today?
- What share of Travel support contacts are debit-without-ticket, and what is the median time to resolution?
- Where must the flow hand off to IRCTC (login, captcha), and how does that shape a voice journey?
- Is post-booking voice support an acceptable first launch, given its lower risk and clearer ROI?

> **Beyond Travel.** The same harness (intent and slot extraction, grounded answers, payment-safe guardrails, in-conversation failure recovery) transfers directly to bill payments, recharges and insurance servicing. The investment compounds across the app.

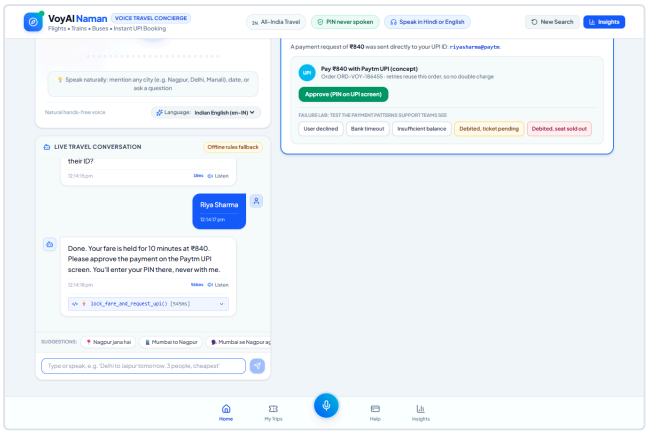
A Appendix: prototype and assumptions

EVIDENCE

A working prototype exists to test the riskiest assumptions: whether Hinglish trip requests can be understood in one turn, and whether an LLM can sit in a payment flow without ever producing a money value. Code: github.com/devanshudhoble/namanai.



"Sabse sasta wala" → cheapest option read back with fare, timing and refund rule before asking for approval.



Fare held for 10 minutes, in-app UPI, and every payment failure pattern reproducible for testing.

What the prototype proves, and what it doesn't

COMPONENT	STATUS
LLM understanding with schema and server validation	WORKING
Context harness: guards, budget, ledger, validator	WORKING
Offline fallback engine and Hinglish date parsing	WORKING
Indic text-to-speech via server	WORKING
Journey analytics, latency p50/p90	WORKING
Train, bus and flight inventory	SANDBOX
UPI, bank callbacks, refunds	SANDBOX

27 automated tests cover the harness, refund maths and every payment-failure journey. A 30-utterance English/Hindi/Hinglish eval set acts as a regression floor; a production set would come from real search and support language.

Assumptions log

ASSUMPTION	HOW TO TEST
Hindi-first users leak most at search and payment	Funnel by app language and city tier
Fear of a wrong booking outweighs price for first-timers	Interviews; review-page dwell time
Debit-without-ticket is a top-5 support driver	Support intent tagging
Trains have the most first-time bookers	First-booking share by mode
Voice won't raise wrong bookings with read-back	Pilot guardrail